

Design and Implementation of a Compiler with Machine-Learning-Guided Optimization Selection

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Abstract—This paper documents the complete design, implementation, and evaluation of MiniLang, a research compiler for a minimal statically typed imperative language. MiniLang was build from scratch in C++17 to support end-to-end study of compiler construction and machine-learning-guided optimization. The system implements a classical multi-phase pipeline:hand-written lexical analysis, recursive-descent parsing, two-pass semantic analysis, lowering to a flat High-level Intermediate Representation(HIR), configurable rule-based optimizations, textual LLVM IR emission, and an ML-ready optimization advisor with evaluation harness.

We describe every compilation phase in implementation-level detail, including token design, abstract syntax tree (AST) structure, semantic rules, HIR instruction set, three optimization passes, LLVM code generation strategy, and a feature-driven advisor that selects optimization pipeline using heuristics, JSON plans or external Python interface. Experimental results in benchmark programs show that optimization reduces the HIR instruction count by up to 63% ($22 \rightarrow 8$ instruction) and LLVM IR size by 41% ($34 \rightarrow 20$ lines) on constant-heavy workloads, while preserving correct runtime behavior(verified by process exit codes.) Minilang is intended as a reproducible research and teaching traditional compiler engineering and modern ML-for-compilers research

Index Terms—Compilers, intermediate representation, LLVM, static analysis, code optimization, machine learning, pass ordering

I. INTRODUCTION

A. Motivation

II. EASE OF USE

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	Table column subhead	Subhead	Subhead
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Fig. 1. Example of a figure caption.

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